Code-Layout, Readability and Reusability

Date	05 July 2026
Team ID	
Project Name	Green-Predict-Pro
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing / tabs used throughout the code	Yes	4-space indentation, PEP 8.
2	Proper File Structure	Files and folders are logically organized	Yes	app.py, /templates, /static, /model, /notebooks.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	e.g., nitrogen_level, rainfall_mm, predicted_crop.
4	Function / Method Names	Functions are descriptively named	Yes	load_model(), predict_crop(), preprocess_input().
5	Code Comments	Inline and block comments explain logic	Yes	Docstrings on every function and route.
6	Modular Design	Code is split into reusable functions / modules	Yes	Separate modules for model, preprocessing and routes.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Common logic refactored into helpers.
8	Error Handling	Exceptions and errors are handled	Yes	try/except with user-

		gracefully		friendly messages.

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High/Medium/Low)
1	preprocess.py	Python	Input scaling for both training and prediction pipelines.	High
2	model_utils.py	Python / scikit-learn	Loads model, returns prediction & confidence – reused by web app and Jupyter notebooks.	High
3	form_validator.py	Python	Reused across all input forms for range validation.	Medium
4	charts.py	Matplotlib / Seaborn	Reusable chart functions for the researcher dashboard.	Medium
5	auth_utils.py	Flask	Login, logout and password hashing helpers reused across routes.	Medium

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Modular, PEP 8 compliant.
Readability	5	Clear naming and docstrings throughout.
Reusability	4	Core ML and utility functions reused across notebooks and web app.
Documentation / Comments	4	Docstrings + README + inline comments.
Overall Score	4.5	Production-ready quality.